Roll No.: _

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech., Missed Mid-Term Examinations – December 2025

Third Semester

Artificial Intelligence & Data Science

23AID202 Introduction to Robotics

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Explain facts pertaining to robotics including history, sub-fields, and applications.
CO02	Explain the elementary concepts required for modelling robotic systems.
CO03	Develop mathematical and mechanistic models for simple robotic systems.
CO04	Use tools such as Robotics toolbox to program and visualize simple robotic systems

Answer all questions

PART I

1. The figure below shows three coordinate frames:

- The world frame {0}
- Frame {A}, obtained by:
  - Translating the world frame by (3,−1)
  - Rotating it by 30° counterclockwise
- Frame {B}, obtained by:
  - Translating the world frame by (7,3)
  - Rotating it by 60° counterclockwise

A point Q (9,4) is represented in the world coordinate frame.

i) Using homogeneous transformation matrices, compute the coordinates of point Q with respect to frame {A}.

Show all steps:

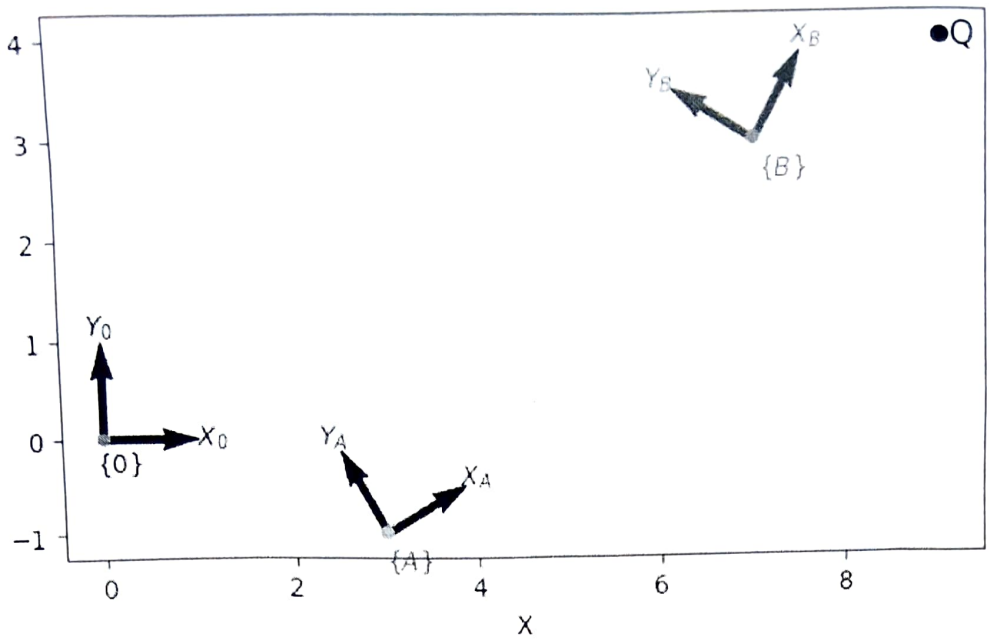
- Rotation matrix
- Homogeneous matrix
- Inverse transform
- Final point in {A}

ii) Using homogeneous transformation matrices, compute the coordinates of point Q with respect to frame {B}.

Show all steps:

- Rotation matrix
- Homogeneous matrix
- Inverse transform
- Final point in {B}

[20][CO03][BTL Level: 4 – Analyze]



2. Using the SCARA kinematic equations, compute the homogeneous transformation matrix 4×4 matrix of the end-effector (base $\rightarrow$*to*$\rightarrow$ tool) for the IRB 920T-6/0.65 robot when the joint values and geometry are:

- Joint variables:
  - $\theta 1 = -45^\circ$
  - $\theta 2 = 60^\circ$
  - $d3 = -120$ mm
  - $\theta 4 = 30^\circ$ (tool rotation about local z)
- Robot geometry (assumed):
  - a1 =0.40 m
  - a2 =0.25 m
  - d0 =0.20 m (height of the base to joint 1 axis)

[20][CO03][BTL Level: 4 – Analyze]

3. A two-wheel differential-drive mobile robot has two identical DC motors mounted on the left and right wheels.
Each wheel has a radius r, and the wheels are separated by a distance L.
The total mass of the robot is m, and the moment of inertia of the robot about its vertical axis passing through its center of mass is Iz .

When torques τL and τR are applied by the left and right wheel motors, the robot experiences linear and angular accelerations.

[10][CO02][BTL Level: 2 – Understand]

****

CO	Marks	BTL	Marks
CO01		BTL 1	
CO02	10	BTL 2	10
CO03	40	BTL 3	40
CO04		BTL 4	
CO05		BTL 5	
CO06		BTL 6	

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create